

TASK – 2

E-COMMERCE SYSTEM – PRODUCT & CATEGORY

1. Aim

To create **CATEGORY** and **PRODUCT** tables for an e-commerce system and perform product operations and category-wise analysis.

2. Category Table Design

Column	Data Type	Constraint
Category_ID	INT	Primary Key
Category_Name	VARCHAR(50)	NOT NULL, UNIQUE

SQL Code

```
CREATE TABLE Category (  
    Category_ID INT PRIMARY KEY,  
    Category_Name VARCHAR(50) NOT NULL UNIQUE  
);
```

3. Product Table Design

Column	Data Type	Constraint
Product_ID	INT	Primary Key
Product_Name	VARCHAR(100)	NOT NULL, UNIQUE
Price	DECIMAL(10,2)	CHECK
Stock_Quantity	INT	CHECK
Category_ID	INT	Foreign Key, NOT NULL

SQL Code

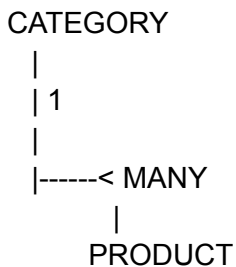
```
CREATE TABLE Product (  
    Product_ID INT PRIMARY KEY,  
    Product_Name VARCHAR(100) NOT NULL UNIQUE,
```

```
Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
Stock_Quantity INT NOT NULL CHECK (Stock_Quantity >= 0),
Category_ID INT NOT NULL,
FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
);
```

4. Relationship

Category (1) → Product (Many)

- One category can contain many products.
- Each product belongs to one category.



5. Insert Category Details

```
INSERT INTO Category VALUES
(1, 'Electronics'),
(2, 'Clothing'),
(3, 'Books'),
(4, 'Home Appliances');
```

6. Insert 10 Product Records

```
INSERT INTO Product VALUES
(101, 'Laptop', 55000.00, 10, 1),
(102, 'Smartphone', 25000.00, 20, 1),
(103, 'Headphones', 2000.00, 30, 1),
(104, 'Keyboard', 1500.00, 25, 1),
(105, 'Mouse', 800.00, 40, 1),
(106, 'T-Shirt', 700.00, 50, 2),
(107, 'Jeans', 1500.00, 30, 2),
(108, 'Novel', 500.00, 15, 3),
```

```
(109, 'Mixer Grinder', 4500.00, 12, 4),  
(110, 'Air Cooler', 9000.00, 8, 4);
```

7. CRUD Operations

CREATE – Add New Product

```
INSERT INTO Product  
VALUES (111, 'Smart Watch', 3500.00, 15, 1);
```

READ – Display Products

```
SELECT * FROM Product;
```

UPDATE – Modify Product Price

```
UPDATE Product  
SET Price = 3000.00  
WHERE Product_ID = 103;
```

UPDATE – Increase Stock

```
UPDATE Product  
SET Stock_Quantity = Stock_Quantity + 20  
WHERE Product_ID = 101;
```

DELETE – Remove Discontinued Product

```
DELETE FROM Product  
WHERE Product_ID = 111;
```

8. Category-wise Product Reports

A. Display All Products Under Each Category

```
SELECT c.Category_Name, p.Product_Name  
FROM Category c  
JOIN Product p  
ON c.Category_ID = p.Category_ID  
ORDER BY c.Category_Name;
```

B. Count Number of Products in Each Category

```
SELECT c.Category_Name, COUNT(p.Product_ID) AS Product_Count
FROM Category c
LEFT JOIN Product p
ON c.Category_ID = p.Category_ID
GROUP BY c.Category_Name;
```

C. Find Highest-priced Product in Each Category

```
SELECT c.Category_Name, MAX(p.Price) AS Highest_Price
FROM Category c
JOIN Product p
ON c.Category_ID = p.Category_ID
GROUP BY c.Category_Name;
```

D. Display Categories Having More Than 5 Products

```
SELECT c.Category_Name, COUNT(p.Product_ID) AS Product_Count
FROM Category c
JOIN Product p
ON c.Category_ID = p.Category_ID
GROUP BY c.Category_Name
HAVING COUNT(p.Product_ID) > 5;
```

E. Find Average Product Price Category-wise

```
SELECT c.Category_Name, AVG(p.Price) AS Average_Price
FROM Category c
JOIN Product p
ON c.Category_ID = p.Category_ID
GROUP BY c.Category_Name;
```

9. Final Submission

1. **Category and Product Table Design**
2. **SQL Table Creation Script**
3. **Sample Product Data**
4. **CRUD Queries**
5. **Category-wise Product Analysis Report**